

Assignment 3 — Matching

Cost-Sharing and the Spending of “Superutilizers”

Choose your method. For this assignment you complete **one** of three options — matching, instrumental variables, or difference-in-differences — whichever best fits the research you care about. All three share the structure below; this file is the **Matching** option. Submit one knitted R Markdown file, acknowledge your group, and include the AI-use statement. **Four blocks:** A and D are your own thinking (most weight), B is AI-assisted execution, C tests whether you catch AI's mistakes. Late penalty 10%/day.

Video (★). The ★ part below requires a video of no more than 3 minutes explaining how you got your answer and what it means.

The data. Use *Dataset 3a_Claims* (in the Assignments folder): spending for individuals in the top 1% of spenders, 2006–2018. Treatment *hdhp* indicates enrollment in a high-deductible health plan; outcome *pay* is total costs. **Context (Finkelstein et al., NEJM 2020):** “superutilizers” are under 0.5% of patients but over 10% of hospital spending; a standard question is whether exposing them to cost-sharing reduces their use.

Research question. Does enrolling superutilizers in a high-deductible plan (*hdhp*) reduce their total spending (*pay*)?

Block A — Setup & Judgment (no code)

Choose a journal you like where a study like this could end up (clinical, econ, sociology, up to you). Using that journal's tone and structure, write the introduction to an interesting paper answering this question. You can limit yourself to 1-2 pages max here (no need to fill all that space either). Some possible questions you could address include:

- a. **Why should anyone care?** In one paragraph, motivate this question from a physician's, a researcher's, and a policy-maker's vantage point. What would each do differently if they knew the answer?
- b. **Anticipate the selection.** What is the underlying comparison in an ideal world, and what do you have in the data? Which confounders can you observe in the data, and which can you not? What are the main identification challenges?
- c. **State the identifying assumption in one sentence.** Complete precisely: “A regression of spending on HDHP enrollment recovers the causal effect only if ...” Then give one reason to believe it and one reason to doubt it *in this setting*.

Block B — AI-Assisted Execution (use AI deliberately)

Use **AI to write the code** — fluent, well-directed use is the goal. For **one** part, paste in the prompt you gave, the raw code returned, and a sentence on what you corrected or redirected (your prompt + critique artifact).

- a. **Clean and choose covariates.** Select covariates you think matter, choose measurement/transformations, and handle missing data. Describe and justify your choices.

- b. **Balance table (before matching).** Build a balance table comparing HDHP vs. non-HDHP enrollees, with two-way t-tests of differences. Are the groups ex-ante comparable? Why would you expect differences?
- c. **Naïve regression.** Regress spending on HDHP enrollment with no matching. Given your balance table, how should these results be interpreted, and how do they likely compare to the true causal effect?
- d. **Matching estimation.** Perform a matching estimation on your covariates. Report the updated balance table for the matched sample and the updated regression. Discuss the key changes and interpret your results.

Block C — The Error Hunt (catch what AI gets wrong)

A student asked an AI assistant to run this analysis and pasted the reply below without checking it. **The code runs without any error message** and the numbers look plausible — but it has **at least three consequential problems** (a mix of coding bugs and interpretation mistakes).

Reproduce it on the real data, then find and fix the problems. For each: (i) what is wrong, (ii) the corrected code or statement, (iii) one sentence on how it changes the conclusion. Flag any extra problem you find that we did not plant.

The AI-generated analysis:

```
library(tidyverse); library(MatchIt); library(fixest)
df <- read_csv("Dataset3a_Claims.csv")

# Match HDHP vs. non-HDHP enrollees on observed characteristics
m <- matchit(hdhp ~ age + sex + region + pay, # covariates to balance on
             data = df, method = "nearest")
md <- match.data(m)

# Effect of HDHP enrollment on spending in the matched sample
est <- feols(pay ~ hdhp, data = md)
summary(est) # default standard errors

# "Because we matched on observed characteristics, this is the causal
# effect of HDHPs on spending, purged of confounding."
```

Block D — Interpret & Defend

- a. ★ **Translate for a decision-maker.** A benefits director asks: “If we move our high-cost members to a high-deductible plan, how much will their spending fall, and how confident are you?” Answer in plain language in your video. Be explicit about whether your estimate is the causal number they want and what would have to be true for it to be.
- b. **Put your results into context.** Assume that you fully believe your results. Write the first two paragraphs of a Discussion section that interprets these results in a way that you think an editor would find compelling. You can compare them to external literature, report clinical/economic significance, or find some other “headline” way to showcase these results.

- c. **Revisit your assumption.** Return to your one-sentence identifying assumption. After matching, are you more or less worried about it? Create and report at least one piece of evidence that you have not already reported (e.g., a sensitivity check) that speaks to whether the conditional-independence assumption is credible here. Why is your evidence imperfect?

Mini Referee Report

Peer review is the part of the job no AI can do for you. AI assistants summarize happily but rarely land the concern that actually threatens a result — and they tend to agree with whatever they are reviewing. That judgment is yours.

Assigned paper: Doyle, Joseph J. (2007). "Child Protection and Child Outcomes: Measuring the Effects of Foster Care." *American Economic Review*.

Read the assigned paper and write a short referee report (about one page, 400–600 words) naming the **two or three MAJOR concerns** — the ones that would change your confidence in the main result — not minor issues of writing. Structure it as: (1) a 2–3 sentence summary of the claim and how it is identified; (2) two or three major concerns, each with why it threatens the central result and what evidence would resolve it; and (3) a verdict naming the single most important revision.

Using AI here. You may use AI to understand the paper, but the concerns must be your own — its first list is usually generic and agreeable. If you used it, note in one line what it suggested and whether you agreed.